

Four-Model vs Three-Model Post-Processing Comparison

Objective

This note summarizes a focused comparison between:

- **4-model candidate**

`embedding + bert + xgb + reranker` thresholds: `consensus=0.0`, `majority\_3=0.0`, `majority\_2=0.8`

- **3-model production-style setups**

all four possible `3 out of 4` model combinations thresholds: `majority\_3=0.5`, `majority\_2=0.5` `no\_majority -> manual`

The comparison was run on both:

- **Post-production split**: `df\_after\_split.csv`
- **Pre-production split**: `df\_before\_split.csv`

The main business question is simple:

Does moving from the current 3-model structure to the 4-model structure improve quality enough to justify the loss in pass-through?

What The Main Fields Mean

- **Anaf pass rate**: share of all rows that pass the `Anaf` post-processing gate.
- **Mishlah pass rate**: share of all rows that pass the `Mishlah` post-processing gate.
- **Joint pass rate**: share of all rows that pass both `Anaf` and `Mishlah`; this is the main pass-through metric.
- **Anaf accuracy on joint passed rows**: `Anaf` correctness measured only on rows that jointly passed.
- **Mishlah accuracy on joint passed rows**: `Mishlah` correctness measured only on rows that jointly passed.
- **Joint exact match accuracy**: share of jointly passed rows where both `Anaf` and `Mishlah` are correct.

In practice, the most important fields to focus on are:

- **Joint pass rate**
- **Anaf accuracy on joint passed rows**
- **Mishlah accuracy on joint passed rows**
- **Joint exact match accuracy**

What Was Compared

For each split, five scenarios were evaluated:

1. **4 models candidate**
2. **3 models: embedding + bert + xgb** (current production trio)
3. **3 models: embedding + bert + reranker**
4. **3 models: embedding + xgb + reranker**
5. **3 models: bert + xgb + reranker**

For the 3-model case, a unanimous `3/3` agreement is treated as `majority\_3`.

Main Results

Post-Production Split

Scenario	Joint pass rate	Anaf acc on joint passed	Mishlah acc on joint passed	Joint exact match
4 models candidate	75.09%	90.09%	89.31%	81.78%
3 models: embedding + bert + xgb	76.67%	88.45%	88.34%	79.61%
3 models: embedding + bert + reranker	72.77%	89.14%	89.08%	80.87%
3 models: embedding + xgb + reranker	81.46%	87.65%	87.21%	77.99%
3 models: bert + xgb + reranker	74.03%	88.96%	88.23%	79.87%

Pre-Production Split

Scenario	Joint pass rate	Anaf acc on joint passed	Mishlah acc on joint passed	Joint exact match
4 models candidate	75.35%	87.47%	86.12%	76.81%
3 models: embedding + bert + xgb	76.45%	85.17%	84.91%	74.02%
3 models: embedding + bert + reranker	72.75%	86.74%	86.66%	76.65%
3 models: embedding + xgb + reranker	81.42%	84.98%	84.36%	73.60%
3 models: bert + xgb + reranker	74.23%	86.46%	86.04%	76.02%

Where To Focus

There are two different winners depending on what is being optimized:

- **Highest volume winner**:

`embedding + xgb + reranker` This setup gives the highest joint pass rate in both splits.

- **Highest quality winner**:

the **4-model candidate** This setup gives the best overall quality profile, especially on `Anaf` joint accuracy and joint exact match.

This means there is no magic solution. The choice is a direct tradeoff:

- more pass-through
- or better quality on the rows that pass

Current Production Trio vs 4-Model Candidate

This is the most important head-to-head comparison.

Post-Production Split

- **Joint pass rate**:

`76.67% -> 75.09%` a drop of **1.57 points**

- **Anaf accuracy on joint passed rows**:

`88.45% -> 90.09%` an improvement of **1.64 points**

- **Mishlah accuracy on joint passed rows**:

`88.34% -> 89.31%` an improvement of **0.97 points**

- **Joint exact match accuracy**:

`79.61% -> 81.78%` an improvement of **2.18 points**

Pre-Production Split

- **Joint pass rate**:

`76.45% -> 75.35%` a drop of **1.10 points**

- **Anaf accuracy on joint passed rows**:

`85.17% -> 87.47%` an improvement of **2.30 points**

- **Mishlah accuracy on joint passed rows**:

`84.91% -> 86.12%` an improvement of **1.22 points**

- **Joint exact match accuracy**:

`74.02% -> 76.81%` an improvement of **2.79 points**

Pre vs Post Production

The key pattern is that the **joint pass rate** is relatively stable before and after split, while the **quality metrics** improve after split.

For the 4-model candidate:

- Joint pass rate:

`75.35%` before split vs `75.09%` after split

- Anaf joint accuracy:

`87.47% -> 90.09%`

- Mishlah joint accuracy:

`86.12% -> 89.31%`

- Joint exact match:

`76.81% -> 81.78%`

For the current production trio:

- Joint pass rate:

`76.45%` before split vs `76.67%` after split

- Anaf joint accuracy:

`85.17% -> 88.45%`

- Mishlah joint accuracy:

`84.91% -> 88.34%`

- Joint exact match:

`74.02% -> 79.61%`

This is consistent with a production bias effect:

- the production-period data is naturally more aligned with the models currently used in production
- analysts may have become more aligned with the model outputs over time
- model-driven workflow effects may also increase agreement after deployment

So the post-production split should be read as slightly favorable to the current live setup.

Bottom Line

The 4-model setup is not a dramatic breakthrough, but it is a real quality upgrade.

A fair summary is:

- the **4-model candidate improves quality by roughly 1 to 3 points**, with the clearest gain showing up in **joint exact match**
- the cost is a **roughly 1 to 1.6 point reduction in joint pass rate** versus the current production trio

If the goal is **maximum pass-through**, the strongest 3-model option is:

- ``embedding + xgb + reranker``

If the goal is **better quality on passed rows**, the strongest option is:

- the **4-model candidate**

So there is no free lunch here:

- **3 models can buy more volume**
- **4 models buy better accuracy**

That is the core decision.